



YOUR TECHNOLOGY & INNOVATION PARTNER

AI-Powered Front-End: From Design to Code

Outline

1. The Role of AI in Frontend
2. Design to Code
3. Live Demo: From Figma design to Flutter code
4. Q&A

The Role of AI in Frontend (1/2)

Why AI Matters in Frontend

- **Speed up development** with UI code generation.
- **Increase accuracy** and reduce human error
- **Bridge design & code** seamlessly
- Assist with architecture & best practices

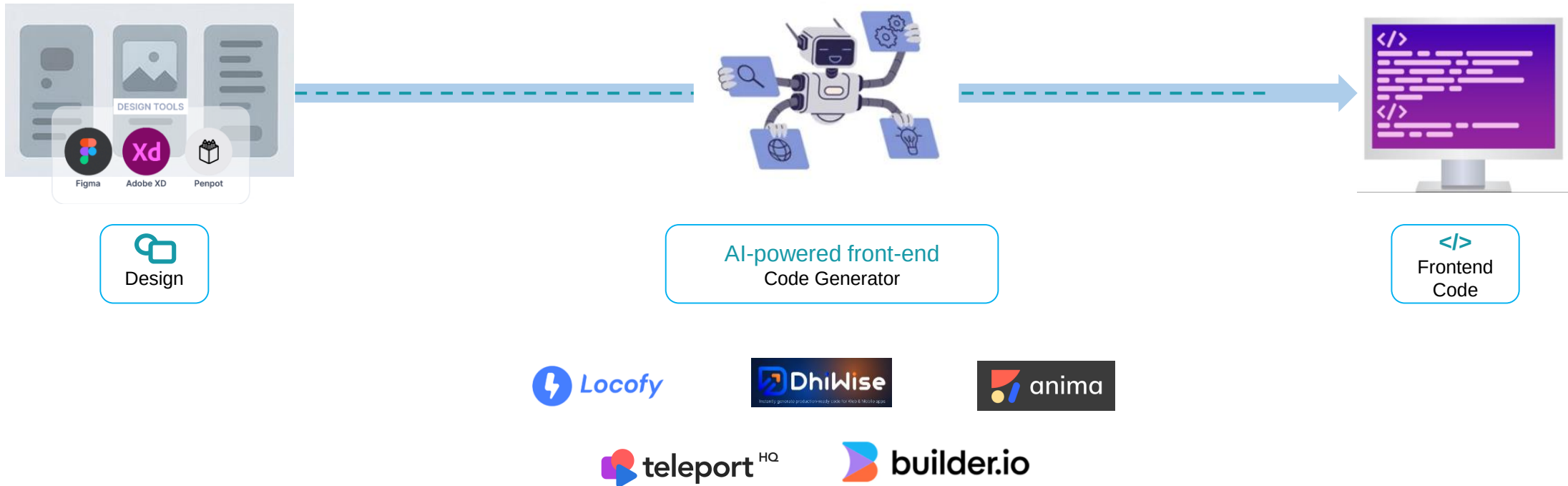
The Role of AI in Frontend (2/2)

AI Use Cases in Frontend

Use Case	How AI Helps	Tools/Examples
Design to Code	Convert Figma/Sketch designs to real code	AI-powered front-end Code Generator, Model Context Protocol Server + AI Assistant
Code Completion	Suggest HTML/CSS/JS/Flutter as you type	GitHub Copilot, Cursor AI
Code Refactoring	Suggest cleaner code and split components	GPT-4, Refact, Cursor AI
Bug Fixing	Explain errors and suggest fixes	GPT-4 Debugger, Cursor AI
Test Generation	Create unit and UI test cases automatically	CodiumAI, GPT-4

Design to Code (1/3)

1. AI-powered front-end Code Generator (1/2)



Design to Code (2/3)

1. AI-powered front-end Code Generator (2/2)

AI Tools	Language Support	AI Prompt-to-Code	Import from design	Quality of code	Rating (Like/Downloads)	Fee
Locofy	React, Next.js, Angular, HTML, Vue, React Native, Flutter	Not supported	Figma, Penpot, Adobe XD, Sketch	Export simple code, can create Component for reuse	4,4k/493k	Monthly 33.3\$/month or 99.9\$/month
Dhiwise (Rocket.new)	Flutter, HTML, NextJs, React	Supported	Figma	Export code clean, responsive, define constants, components	1.2k/125k	Monthly 25-50-100\$/month or Yearly 20-40-80\$/month
Builder.io	React, HTML, CSS, Tailwind, Angular, Vue, React Native, Svelte, SwiftUI, Flutter	Supported	Figma	Export simple code	17.8k/1m	24\$/ month
TeleportHQ	React, Next.js, Vue, Nuxt, Angular, HTML, UIDL	Not supported	Figma	Export simple code, can create Component for reuse	2.9k /346k	Monthly 18\$/month or Yearly 9\$/month
Anima	React, HTML, CSS, Tailwind	Supported	Figma, Adobe XD, Sketch	Export clean code, define component for reuse	13.2k/1.5m	20-40-150\$/month

Design to Code (3/3)

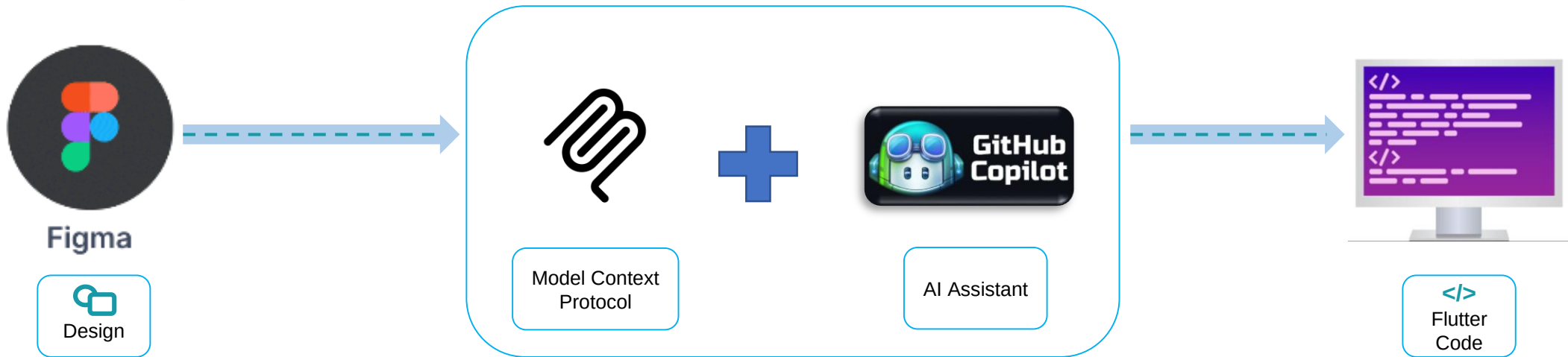
2. Model Context Protocol Server + AI Assistant



Model Context Protocol: It is an open standard that enables developers to build connections between their data sources and AI-powered tools, allowing those tools to understand every detail of design files. MCP solves the problem of AI struggling to accurately convert UI designs into code when only screenshots are available.

- **Figma-Context-MCP:** It's an MCP server that allows AI-powered coding tools (Cursor, Github Copilot, Claude...) to access Figma Design data with this Model Context Protocol server. [@Link](#)
- **Figma Dev Mode MCP Server:** brings Figma directly into your workflow by providing important design information and context to AI agents generating code from Figma design files. [@Link](#)

Live Demo: From Figma design to Flutter code



Model Context Protocol is used for demo: **Figma-Context-MCP**

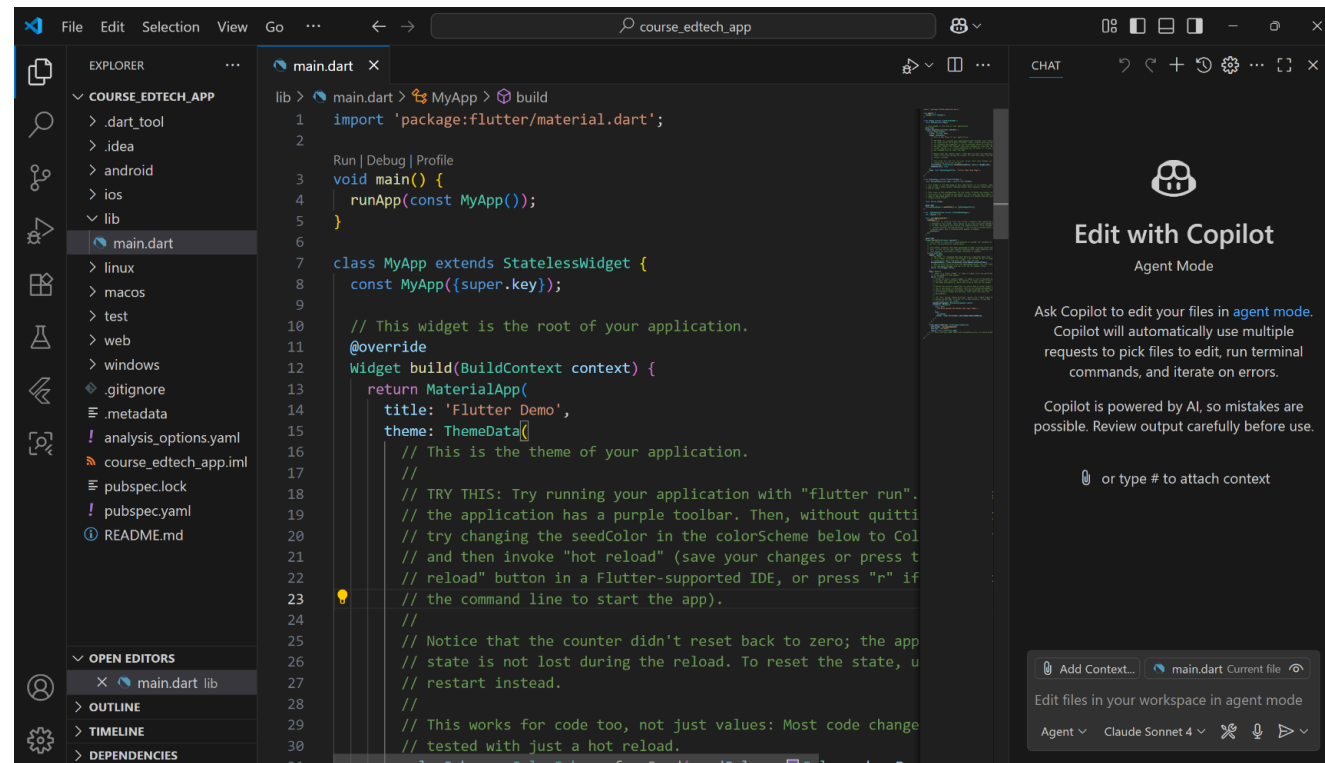
Core Value of **Figma-Context-MCP**

- Allows AI Assistant to directly read Figma design properties (colors, fonts, sizes, components, etc.)
- Provides a standardized interface so AI Assistant interpret design data consistently
- Supports automatic export of assets (images, icons)
- Sends structured data to improve accuracy of AI-generated code

Live Demo: From Figma design to Flutter code

1. Development Environment Setup

- Visual Studio Code
- Flutter
- Github Copilot



Live Demo: From Figma design to Flutter code

2. Add Figma-Context-MCP Server into Github Copilot

2.1 Get Figma API Token: Log in to Figma, go to Settings > Security, and generate a personal access token

Generate new token

A personal access token gives you access to the Figma API as if you were a single user.

AI Github Copilot

Expiration

30 days

This token will expire on August 28, 2025.

Scopes

Comments

Read, post, and delete comments in files

No access

Current user

Read the current user's name, email, and profile image

No access

Dev resources

Read and write to dev resources in files

Read-only

File content

Read the contents of and render images from files

Read-only

File metadata

Read metadata of files

Read-only

File versions

Read-only

Cancel Generate token

Account Community Notifications Security

Personal access tokens

Personal access tokens allow you to access your own data via the API. Do not give out your personal access tokens to anybody who you don't want to access your files.

Generate new token

AI Github Copilot

figd_-oC8

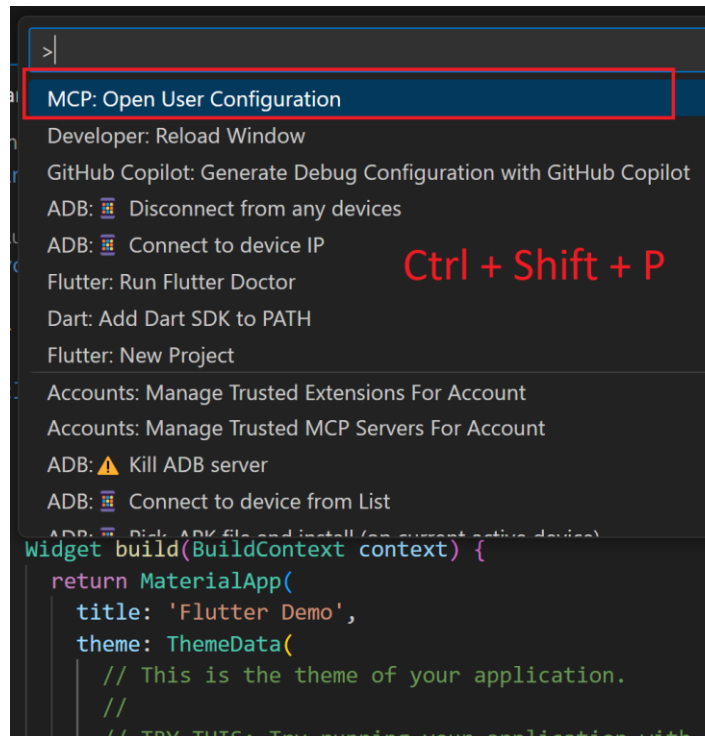
Copy this token. This is your only chance to do so!

Live Demo: From Figma design to Flutter code

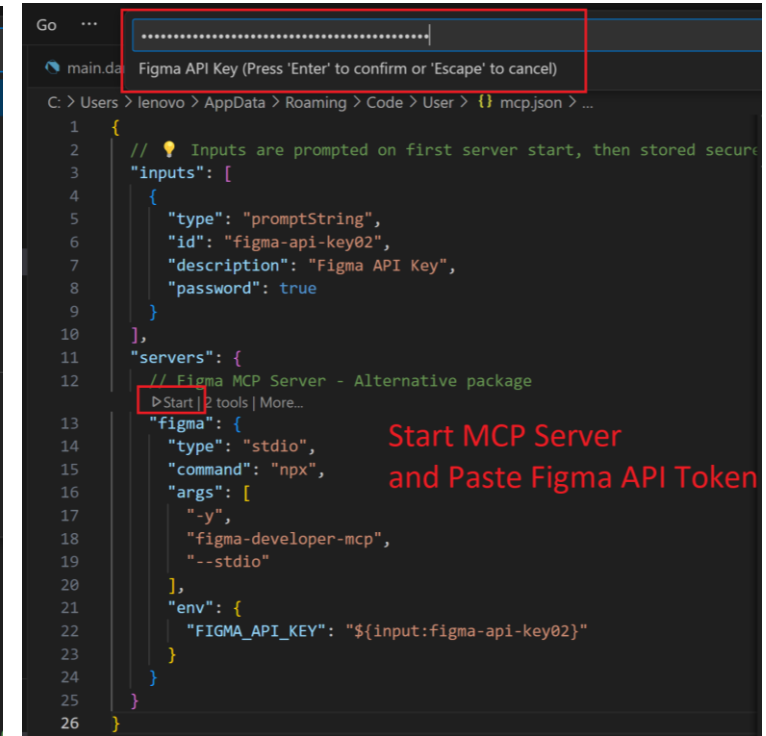
2. Add Figma-Context-MCP Server into Github Copilot

2.2 Configure Figma-Context-MCP into VS Code:

- Open MCP User Configuration
- Add Figma-Context-MCP Server info
- Start the Server and input Figma API token



The screenshot shows the VS Code Command Palette with the search bar containing '>'. The command 'MCP: Open User Configuration' is highlighted. Other visible commands include 'Developer: Reload Window', 'GitHub Copilot: Generate Debug Configuration with GitHub Copilot', 'ADB: Disconnect from any devices', 'ADB: Connect to device IP', 'Flutter: Run Flutter Doctor', 'Dart: Add Dart SDK to PATH', 'Flutter: New Project', 'Accounts: Manage Trusted Extensions For Account', 'Accounts: Manage Trusted MCP Servers For Account', 'ADB: Kill ADB server', 'ADB: Connect to device from List', and 'ADB: Disconnect from any devices'. A red text overlay 'Ctrl + Shift + P' is positioned to the right of the command list.



The screenshot shows the VS Code editor with the 'mcp.json' file open. The file contains a JSON configuration for the Figma MCP Server. A red box highlights the 'inputs' section, which includes a 'figma-api-key02' input. A red text overlay 'Start MCP Server and Paste Figma API Token' is positioned to the right of the configuration. The configuration is as follows:

```
{
  "inputs": [
    {
      "type": "promptString",
      "id": "figma-api-key02",
      "description": "Figma API Key",
      "password": true
    }
  ],
  "servers": {
    // Figma MCP Server - Alternative package
    "figma": {
      "type": "stdio",
      "command": "npx",
      "args": [
        "-y",
        "figma-developer-mcp",
        "--stdio"
      ],
      "env": {
        "FIGMA_API_KEY": "${input:figma-api-key02}"
      }
    }
  }
}
```

For more details, see the [Figma-Context-MCP official site](#)

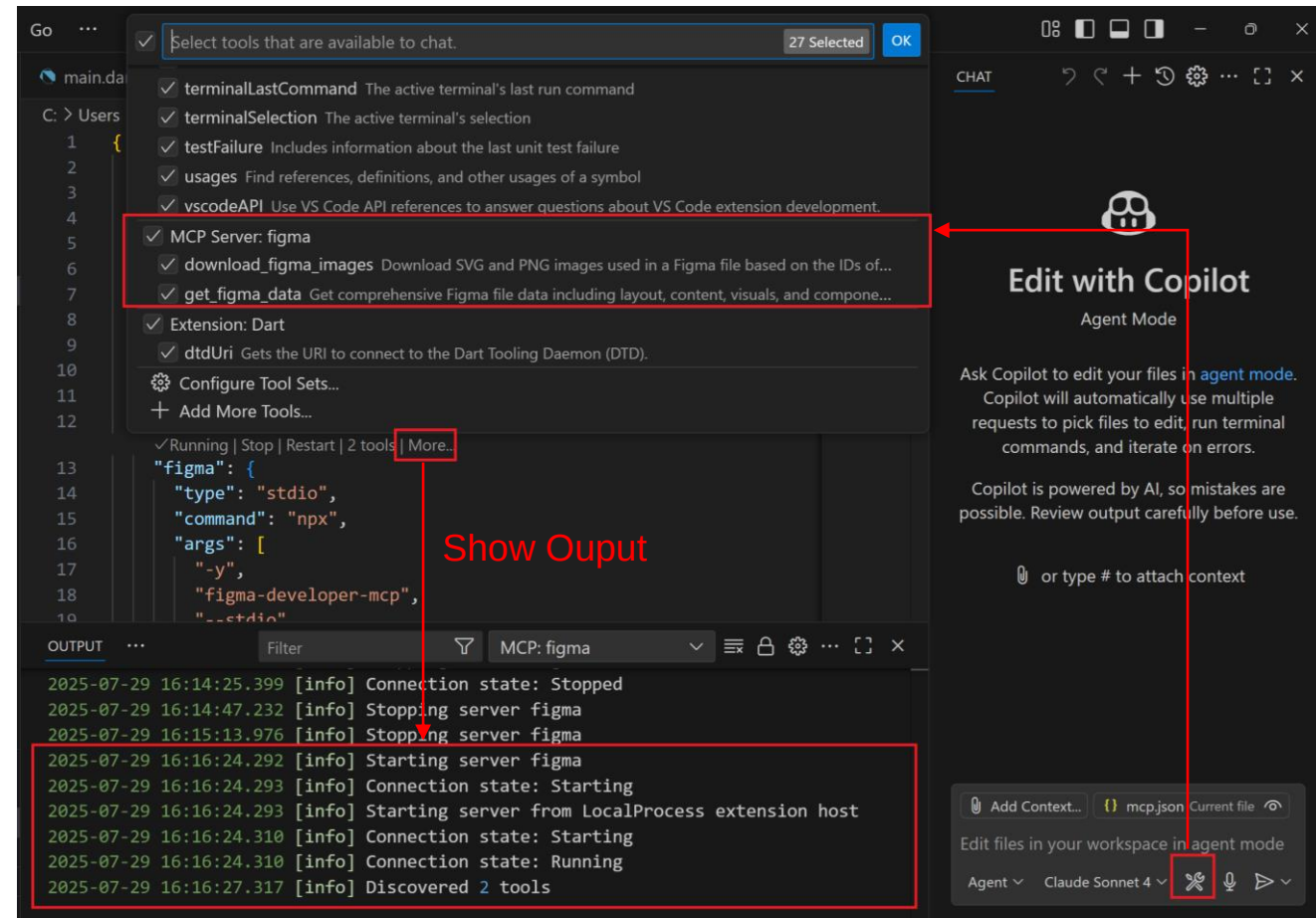
Live Demo: From Figma design to Flutter code

2. Add Figma-Context-MCP Server into Github Copilot

2.2 Configure Figma-Context-MCP into VS Code:

- After starting MCP Server successfully, It will show 2 tools in Copilot – Configure Tools
- **download_figma_images**: Download SVG and PNG images used in a Figma file based on the IDs of image or icon nodes
- **get_figma_data**: Get comprehensive Figma file data including layout, content, visuals, and component information

Now, the Github Copilot is able to access and read necessary data in Figma UI Design file.



Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.1 Copy the Figma Design link

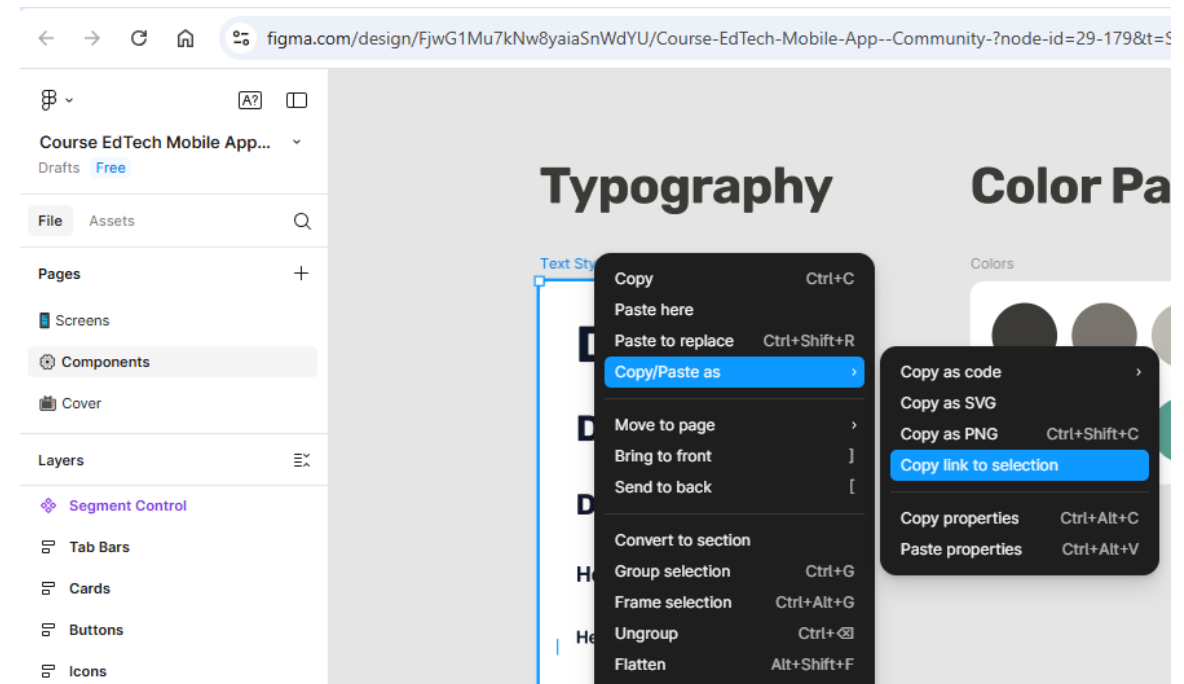
The Figma UI Design is used in this demo: [Course EdTech Mobile App](https://www.figma.com/community/file/928695975857619865/course-edtech-mobile-app)

<https://www.figma.com/community/file/928695975857619865/course-edtech-mobile-app>

In Figma: Select the target frame, right click, and chose
“Copy link to selection”

The copied link includes:

- The Figma file ID
- The node ID (a unique ID of the selected frame or object)



Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.2 Prompt to generate Theme, Style and Color Palettes

- Copy Typography link from UI Design
- Copy Color Palettes link from UI Design

Example Prompt:

You are a senior Flutter developer. Create Theme for project with Text Style and color Palette with the following EXACT requirements

1. Figma Design Links:

- Text Style: Extract from **[Typography link]**
- Color Palette: Extract from **[Color Palettes link]**

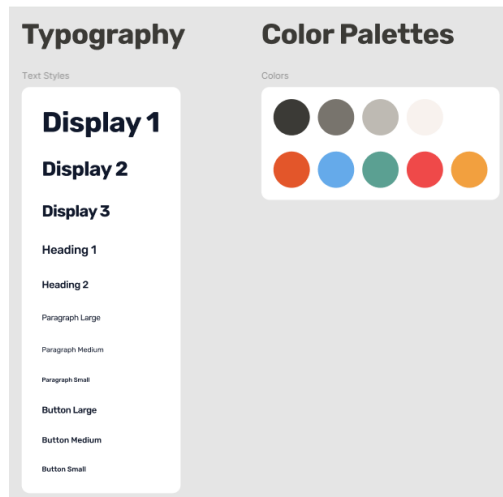
2. Requirements:

- Use Flutter best practices
- Small and Custom reusable components
- Make it responsive for all screen sizes
- Clean, maintainable code structure
- Show Theme Showcase screen with Color Palette, TextStyle interactive examples

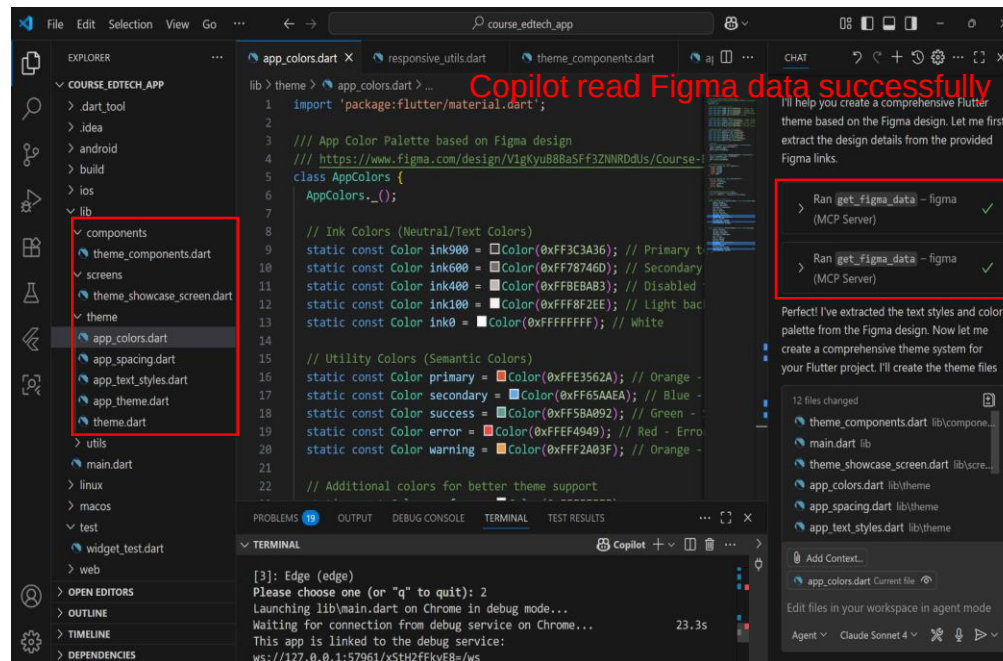
Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

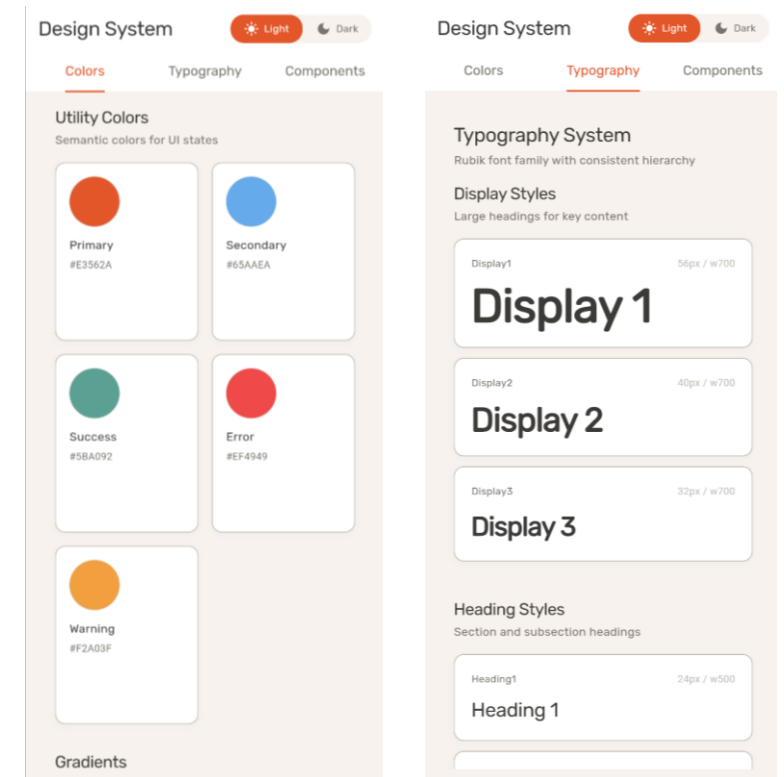
3.2 Prompt to generate Theme, Style and Color Palettes



UI Design



Source code is created by Copilot



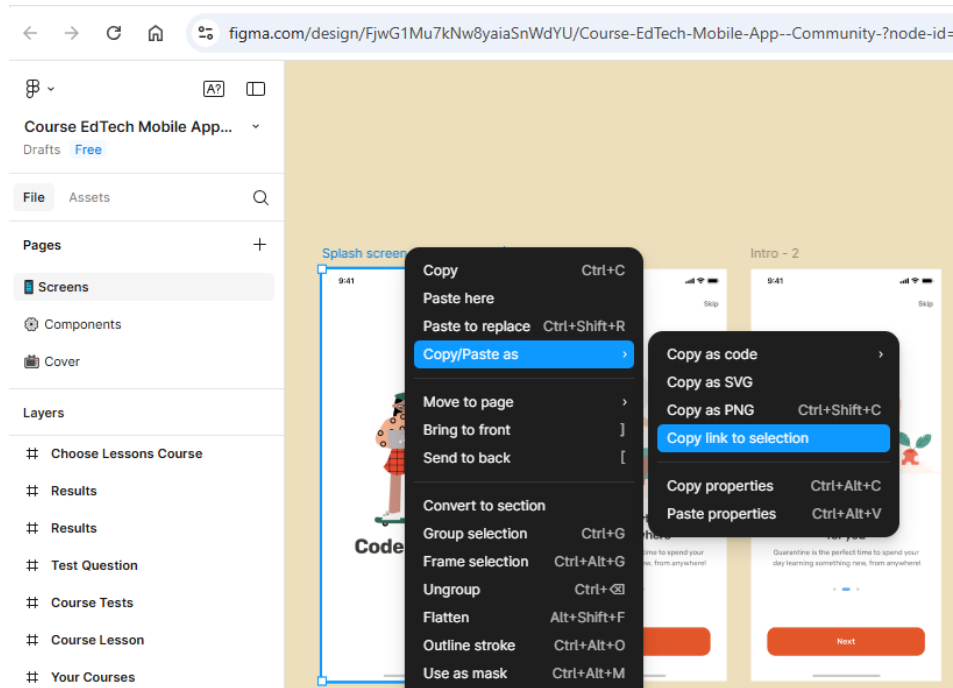
AI generated code result

Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.3 Prompt to generate Splash Screen Source Code

- Copy Splash screen links from UI Design



Example Prompt:

You are a senior Flutter developer. Create a Flutter splash screen with the following EXACT requirements:

1. Figma Design Links:

- Splash Screen: **[Copied link]**,

2. Requirements:

- Auto-navigate to Theme Showcase Screen after exactly 3 seconds
- Download and use images and icons from figma design
- Use Flutter best practices
- Use existing theme system (Text Style, Color)
- Handle safe areas properly
- Clean, maintainable code structure

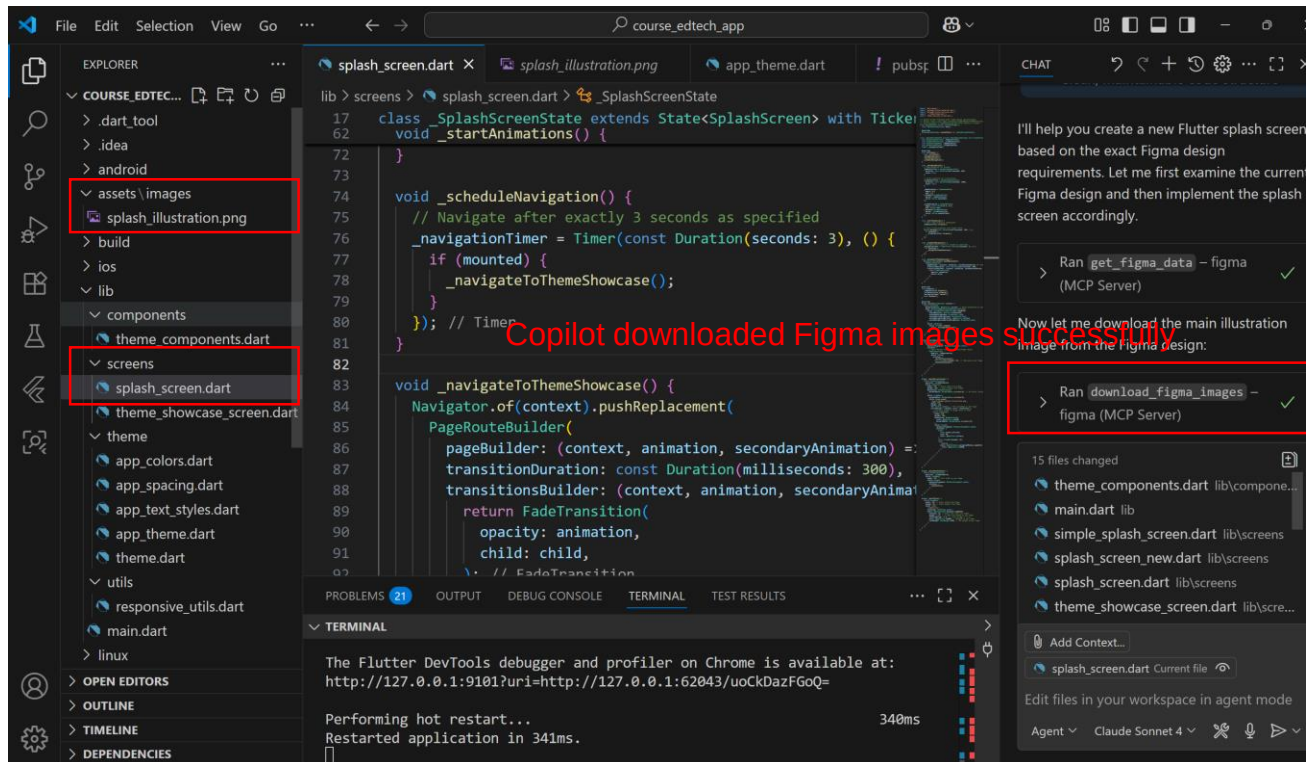
Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

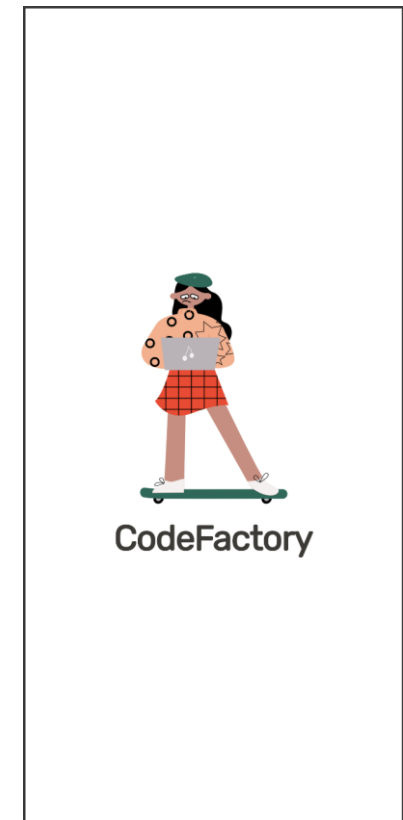
3.3 Prompt to generate Splash Screen Source Code



UI Design



Source code is created by Copilot



AI generated code result

Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.4 Prompt to generate On-boarding Screen Source Code

Example Prompt:

You are a senior Flutter developer.

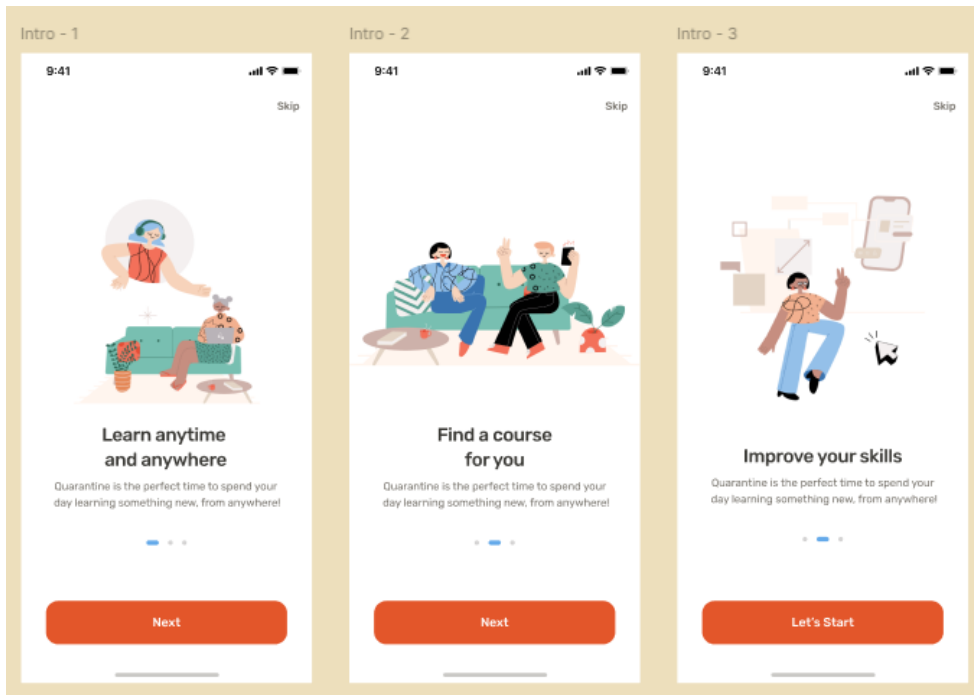
Create a Onboarding screen with the following EXACT requirements:

1. Figma Design Links:

- Intro 1: **[Copied Link]**
- Intro 2: **[Copied Link]**
- Intro 3: **[Copied Link]**

2. Requirement

- Combine Intro 1, Intro 2, Intro 3 into one Onboarding Screen.
- Screen flow: Splash Screen -> Onboarding Screen -> ThemeShowcaseScreen
- Vertically centered on screen
- Single page indicator instance
- Download and use images and icons from figma design
- Use Flutter best practices
- Use existing theme system (Text Style, Color)



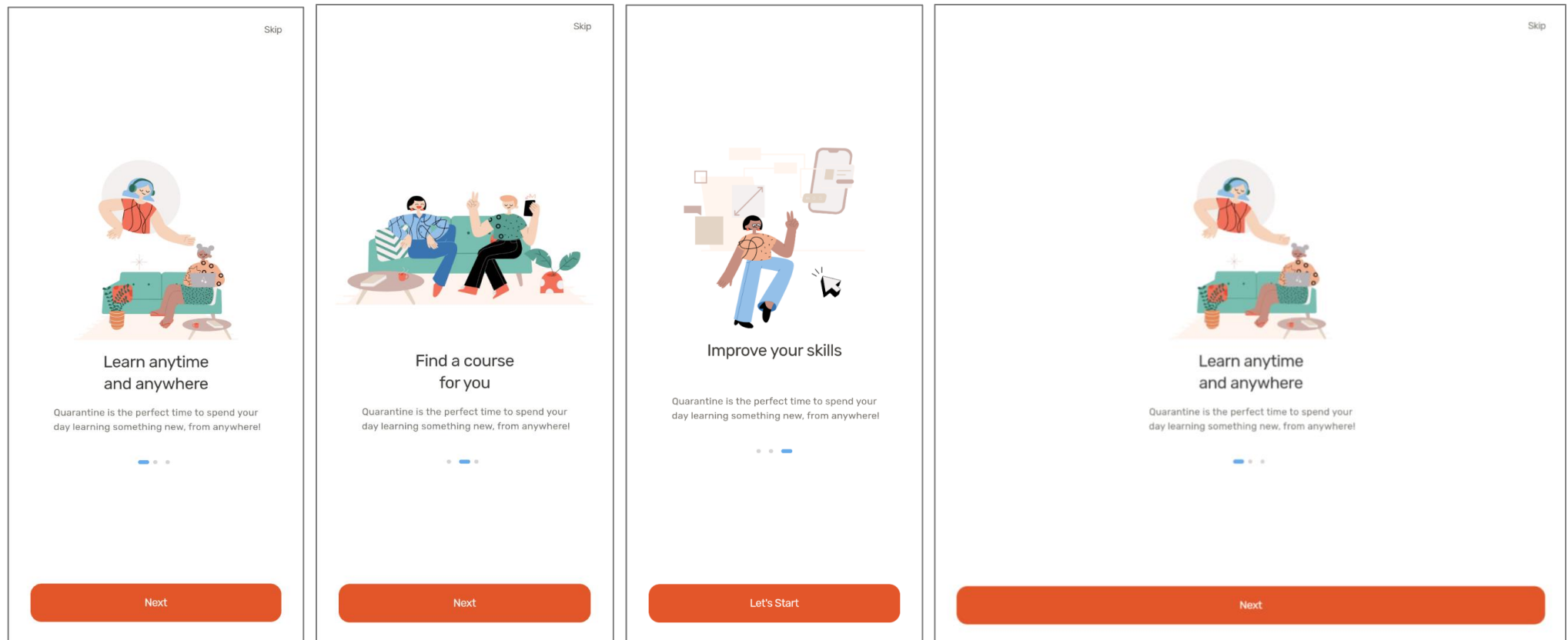
UI Design

Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.4 Prompt to generate On-boarding Screen Source Code

First Result:



Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.4 Prompt to generate On-boarding Screen Source Code

First Result:

The overall structure is clear and correct.

However, a few fine details don't exactly match the design.

- No "Skip" button on the final onboarding screen
- Page indicator position is different on the last onboarding screen
- Font weight is not correctly set to bold
- Button width is not properly constrained

❖ *We can keep refining the output by prompting the AI until the result meets our expectations.*

Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.4 Prompt to generate On-boarding Screen Source Code

Keep prompting

- Show Skip button to appear on all onboarding pages
- Update the onboarding screen to use the exact Figma specifications with proper theme system
- Constraint width of bottom button
- Layout Structure: Use this exact hierarchy

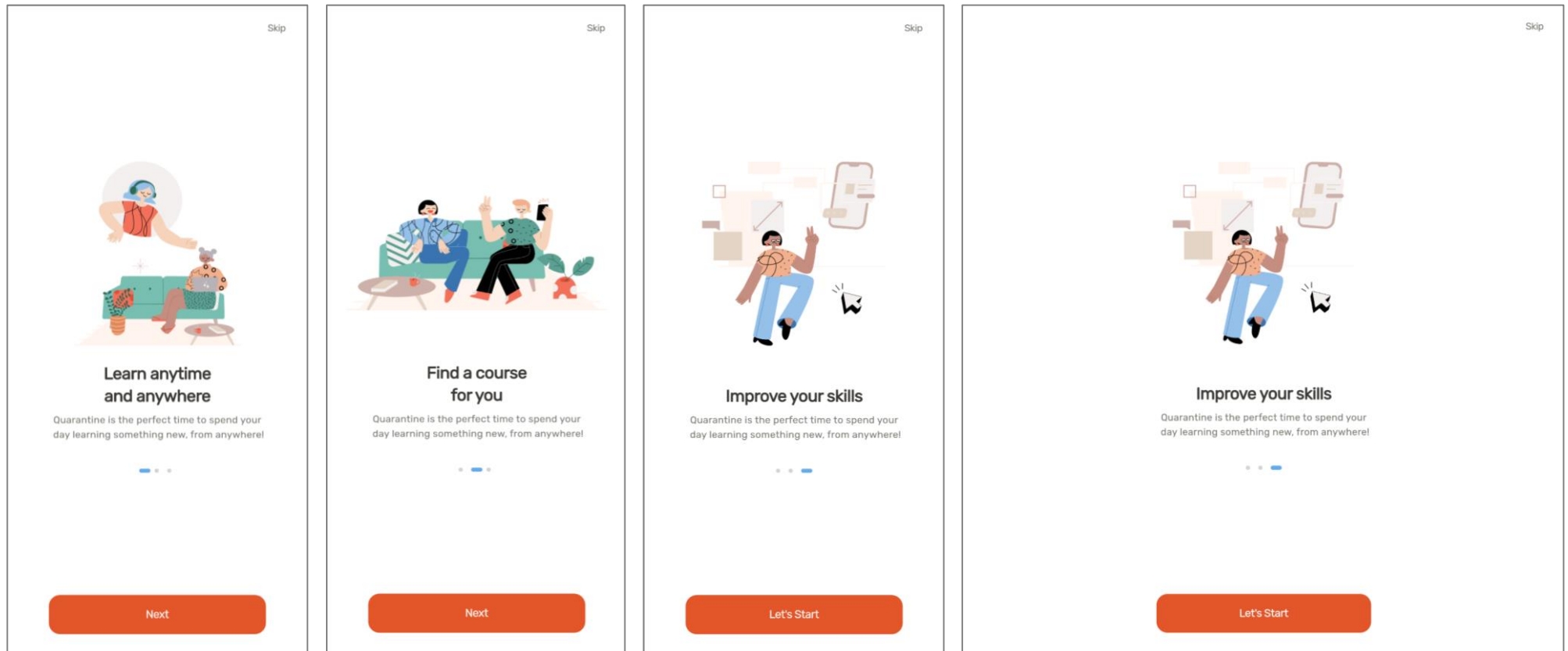
```
Column(  
  Skip Button  
  Spacer() // flexible space above  
  Column(  
    SizedBox(height: 424) //containing PageView  
    SizedBox(height: 16) // exact spacing  
    PageIndicator  
  )  
  Spacer()  
  Bottom Button  
)
```

Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.4 Prompt to generate On-boarding Screen Source Code

Second Result:



Live Demo: From Figma design to Flutter code

3. Prompt to generate source code from Figma Design

3.5 Limitations & Human Review

Using **Figma + Figma-Context-MCP + AI assistants (like GitHub Copilot)** to generate code is a powerful workflow. However, the result often varies from run to run and doesn't exactly 100% match the Figma design

- AI depends on context and prompt quality
- Figma context is not fully translated to code
- AI doesn't fully understand design intent
- AI is non-deterministic

❖ **AI is a productivity and development booster — not a replacement for developers. Manual code review remains essential.**

Q&A



THANK YOU!

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